

Speaker Notes

What to Say for Each Slide

Case Study 5 — Multiple Regression in Biology & Analytics

STAT 482 | Mohammed bin Salem | April 1, 2026

How to use this document:

Read the words in each section out loud when you show that slide.

The blue tips tell you what to do or point at on the screen.

Slide 1 — Title Slide

💡 *Just stand up, smile, and say this while your title slide is on the screen.*

"Hello everyone.

My name is Mohammed bin Salem, and today I will present my Case Study 5 for STAT 482.

The topic is: How do environmental conditions — like temperature, wind, and humidity — affect how big a forest fire gets.

To do this, I used a real dataset from Portugal that has information about 517 forest fires.

The method I used is called Multiple Linear Regression.

Let me start with the data."

Slide 2 — What's the Data?

💡 *Point at the 4 coloured boxes at the top when you say the numbers.*

"So, what is the dataset?

It is called the Forest Fires dataset, from the UCI Machine Learning Repository.

It has 517 rows — each row is one fire event in Montesinho Park in Portugal.

There are 8 predictor variables — these are the things we use to try to predict the fire size.

The good news is: there are zero missing values. The data is clean.

The average burned area is 12.85 hectares — but this number is a bit misleading because a few very large fires pull the average up.

Now, the 8 variables on the left — four of them are fire weather codes called FFMC, DMC, DC, and ISI. These measure things like how dry the soil is or how fast a fire might spread.

The other four are simple weather measurements: temperature, humidity, wind speed, and rainfall.

And our response variable — what we are trying to predict — is the burned area in hectares."

Slide 3 — Exploratory Data Analysis

 *Point at the left charts when talking about the transformation, then point at the table on the right for correlations.*

"Before we build the model, we need to understand the data.

When I looked at the burned area variable, I found a big problem — the distribution is very skewed.

You can see in the left chart: almost all fires burned close to zero hectares, but a few fires burned over 1,000 hectares. This is called a right-skewed distribution, and it is not good for linear regression.

So I applied a log transformation — I replaced area with $\log(\text{area} + 1)$. You can see in the right chart that the distribution looks much more balanced now.

I also checked the correlations between each variable and the response.

As you can see in the table on the right — no variable has a strong correlation with the burned area. The highest value is only 0.07.

This already tells us something important: predicting fire size from weather alone is going to be difficult.

But let's see what the regression model says."

Slide 4 — Full Regression Model Results

 Point at the green equation bar at the top. Then point at the "wind" row in the table — it is highlighted in orange. Then point at the Adjusted R^2 value on the right.

"Now let's look at the regression results.

I ran a full model with all 8 predictor variables. The equation is shown at the top of the slide.

If we look at the coefficient table, we can see the estimate, standard error, and p-value for each variable.

The most important thing to look at is the p-value column. A p-value below 0.05 means the variable is statistically significant.

Only one variable is significant — and that is wind speed. Its p-value is 0.039, which is just below 0.05.

The coefficient for wind is positive — 0.076 — which means: when wind speed goes up, the burned area also goes up. This makes biological sense, because wind helps fire spread faster.

Now look at the model fit on the right side. The Adjusted R-squared is only 0.004 — that means the model explains less than 1% of the variation in fire size.

This is a very weak model — but it is an honest result. Weather conditions alone are not enough to predict how big a fire gets."

Slide 5 — Checking Model Assumptions

💡 *Point at the VIF chart on the left, then the four coloured test cards on the right.*

"Before we trust the model, we need to check that it follows the rules of linear regression.

There are four main checks.

First: Multicollinearity. This means — are the predictor variables too correlated with each other?

I used a test called VIF — Variance Inflation Factor. The chart on the left shows the VIF for each variable. All values are below 5, which is the safe limit. So multicollinearity is not a problem here.

Second: Constant error variance — this means the spread of errors should stay the same across all values.

I used the Breusch-Pagan test. The result is $BP = 11.27$, $p\text{-value} = 0.187$. Because p is greater than 0.05, we pass this test. No problem.

Third: Normality of residuals. I used the Shapiro-Wilk test. The result is $p\text{-value}$ less than 2.2 times 10 to the power of negative 16 — so we fail this test. The residuals are not perfectly normal.

However, this is expected with fire data, because a few extreme fires make the tail very heavy. With 517 observations, our regression results are still valid.

Fourth: I visually checked the residuals vs fitted plot. It shows a slight fan shape, which means the error spread is not perfectly even. But the formal test passed, so we are okay."

Slide 6 — Which Model Is Best?

 Point at the bar chart and mention which bar is lowest. Then point at the teal "Best Model" box on the right. Then point at the comparison table at the bottom.

"I did not just build one model. I compared four different models to find the best one.

To compare models, I used AIC — Akaike Information Criterion. Lower AIC means better model.

As you can see in the bar chart, the Stepwise model has the lowest AIC — 1816.

What is stepwise regression? It is a method that automatically selects the best variables to keep in the model. It removed variables that were not helping, and kept only three: DMC, RH, and wind.

The equation is on the right side of the slide.

In the table at the bottom, you can see all four models compared. The stepwise model has the best AIC and BIC — so it is the winner.

The Time Controls model — which added month and day — had the best Adjusted R-squared at 0.021. But it was penalized for having too many extra variables, so its AIC and BIC were worse.

So for simplicity and fit, the stepwise model with three variables is the best choice."

Slide 7 — How Well Does It Predict?

 Point at the RMSE and MAE cards. Then point at the bullet points on the right side.

"Now we need to check — does the model actually work on new data?

I split the data: 80% for training the model, and 20% for testing it.

That means I trained on 416 observations and tested on 101 observations.

After making predictions on the test set, I got two error values:

RMSE — Root Mean Squared Error — is 1.71. And MAE — Mean Absolute Error — is 1.30. Both are on the log scale.

To understand what this means in practice: an MAE of 1.30 on the log scale is roughly equal to an error of about 3.7 hectares on the original scale. That is quite a lot of uncertainty.

The training model also confirms that wind is the only significant predictor, with a p-value of 0.037.

So the model is honest — it is not pretending to be better than it is. Weather data alone is limited."

Slide 8 — Conclusions & Takeaways

 *Read each numbered point calmly. Pause between points. This is your last slide — end with confidence.*

"Okay, let me summarize what I found.

Number one: Wind speed is the most important variable. It is the only predictor that is statistically significant across all models.

Number two: The stepwise model — with DMC, RH, and wind — is the best model. It has the lowest AIC and BIC values.

Number three: There is no multicollinearity problem. All VIF values are below 5.

Number four: The formal diagnostic tests mostly pass. The error variance test is fine. Normality fails, but this is expected with fire data and does not break our results.

Number five: The model has real limitations. RMSE is 1.71 and MAE is 1.30. To predict fire size better, we would need to add spatial data — like topography and vegetation type.

In conclusion, this analysis shows that wind is the key environmental driver of fire spread. And multiple regression is a useful tool — but also honest about what weather data can and cannot tell us.

Thank you. I am happy to answer any questions."

General Presentation Tips

- Speak slowly. Do not rush.
- Look at the audience sometimes, not just the screen.
- If you forget something, it is okay. Just move to the next sentence.
- You can say "as you can see here" and point at the slide — that buys you time.
- Practice reading this document out loud at least twice before the real presentation.
- It is okay to hold this document in your hand or have it on a tablet.